Status	Finished
Started	Monday, 21 July 2025, 3:55 PM
Completed	Monday, 21 July 2025, 5:30 PM
Duration	1 hour 35 mins
Grade	**5.00** out of 10.00 (**50**%)

Question 1

Correct

Mark 5.00 out of 5.00

Given an integer array arr[ ], and an element key to be found.

Write a function linear_search to find the element in the array. The function should return the index of the key if the key is found, otherwise -1.

Input

three lines of Input: i) N (size of Array) ii) Elements of array separated by comma, iii) key to be found

where

$$1 <= N \leq 1000, \ -10000 \leq key \leq 10000, \ -10000 \leq A[i] \leq 10000$$

Example 1:

Input:

5 (N)

4, 1, 3, 9, 7 (Array Elements)

9 (key)

Output:

3

Example 2:

Input:

10

10, 9, 8, 7, 6, 5, 4, 3, 2, 1

13

Output:

-1

For example:

Input	Result
5 4, 1, 3, 9, 7 9	3
10 10, 9, 8, 7, 6, 5, 4, 3, 2, 1 13	-1

Answer: (penalty regime: 0 %)

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```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
int LinearSearch(int arr[] , int n , int key){
    for(int i=0;i<n;i++){
        if(arr[i]==key){
            return i ;
        }
    }
    return -1;
}
int main(){
    int N;
    scanf("%d", &N);
    getchar();
    int arr[N];
    char line[1000];
    fgets(line , sizeof(line),stdin);
    line[strcspn(line, "\n")]= 0;
    char * token = strtok(line, ",");
    int i = 0;
    while(token != NULL && i<N){
        arr[i] = atoi(token);
        token = strtok (NULL, ",");
        i++;
    }
    int key;
```

	Input	Expected	Got	
✓	5 4, 1, 3, 9, 7 9	3	3	✓
✓	10 10, 9, 8, 7, 6, 5, 4, 3, 2, 1 13	-1	-1	✓
✓	30 198, 227, 216, 798, 652, 329, 287, 80, 381, 299, 585, 34, 422, 841, 640, 389, 865, 822, 367, 854, 196, 899, 842, 953, 338, 52, 436, 417, 829, 119 829	28	28	✓
✓	4 349, 392, 196, 449 551	-1	-1	✓
✓	17 623, 544, 755, 998, 705, 406, 12, 854, 886, 909, 631, 511, 887, 868, 518, 610, 820 182	-1	-1	✓

Passed all tests! ✓

▸ **Show/hide question author's solution (C)**

Correct

Marks for this submission: 5.00/5.00.

Question **2**

Incorrect

Mark 0.00 out of 5.00

Given an integer array arr[ ], and an element key to be found.

Write two functions:

i) **check_sorted_array** to check whether given array is sorted or not. If the Array is sorted, print 1 and call binary_search; otherwise return **405 ,** and print "**Error 405:: Wrong I/P**" in the main function and don't call binary_search function.

ii) binary_search to find the element in the array. The function should return the index of the key if the key is found, otherwise -1.

Input

three lines of Input: i) N (size of Array) ii) Elements of array separated by comma (elements are sorted and in ascending order) iii) key to be found

where

$$1 <= N \leq 1000, \ -10000 \leq key \leq 10000, \ -10000 \leq A[i] \leq 10000$$

Example 1:

Input:

5 (N)

1, 3, 4, 7, 9 (Array Elements)

7 (key)

Output:

3

Example 2:

Input:

10

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

13

Output:

-1

For example:

Input	Result
5 1, 3, 4, 7, 9 7	1 3
10 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 13	1 -1

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
int check_sorted_array(int arr[] ,int size){
    for(int i=0;i<size-1;i++){
        if(arr[i]>arr[i+1]){
            printf("Error 405::  Wrong I/P\n");
            return 405 ;
        }
    }
    return 1;
}
int binary_search(int arr[] , int size , int key){
    int low = 0;
    int high = size - 1;
    while(low<= high){
        int mid = low + (high-low) / 2;
        if(arr[mid]==key){
            return mid;
        }else if (arr[mid]<key){
            low= mid + 1;
            }else{
                high  = mid-1;
            }
        }
        return -1;
    }
int main(){
    int N;
    if (scanf("%d", &N) !=1){
        fprintf(stderr , "Error reading array size. \n");
        return 1;
    }
    while (getchar() != '\n');
    int *arr = (int *) malloc(N * sizeof(int));
    if(arr== NULL){
        fprintf(stderr, "Memory allocation failed.\n");
        return 1;
    }
    char line[1000];
    if (fgets(line , sizeof(line),stdin)==NULL){
        fprintf(stderr, "Error reading array element.\n");
```

Syntax Error(s)

```
__tester__.c: In function ‘main’:
__tester__.c:51:41: error: comparison between pointer and integer [-Werror]
   51 |         if(sscanf(token , "%d", &arr[i] !=1)){
      |                                         ^~
__tester__.c:51:29: error: format ‘%d’ expects argument of type ‘int *’, but argument 3 has type ‘int’ [-
Werror=format=]
   51 |         if(sscanf(token , "%d", &arr[i] !=1)){
      |                            ~^   ~~~~~~~~~~~
      |                             |           |
      |                             int *       int
__tester__.c: At top level:
__tester__.c:60:1: error: expected identifier or ‘(’ before ‘if’
   60 | if(scanf("%d", &key)!=1){
      | ^~
__tester__.c:65:5: error: expected identifier or ‘(’ before ‘if’
   65 |     if(check-sorted_array(arr, N)==1){
      |     ^~
__tester__.c:70:5: error: data definition has no type or storage class [-Werror]
   70 |     free(arr);
      |     ^~~~
__tester__.c:70:5: error: type defaults to ‘int’ in declaration of ‘free’ [-Werror=implicit-int]
__tester__.c:70:5: error: parameter names (without types) in function declaration [-Werror]
__tester__.c:70:5: error: conflicting types for ‘free’; have ‘int()’
In file included from __tester__.c:3:
/usr/include/stdlib.h:555:13: note: previous declaration of ‘free’ with type ‘void(void *)’
  555 | extern void free (void *__ptr) __THROW;
      |             ^~~~
__tester__.c:71:5: error: expected identifier or ‘(’ before ‘return’
   71 |     return 0 ;
      |     ^~~~~~
__tester__.c:73:1: error: expected identifier or ‘(’ before ‘}’ token
   73 | }
      | ^
cc1: all warnings being treated as errors
```

▸ **Show/hide question author's solution (C)**

Incorrect

Marks for this submission: 0.00/5.00.